

MS4215: Advanced Data Analysis

Week 1B: Revision of Statistical Concepts

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Learning outcomes

- Classify a variable as nominal, ordinal, discrete, or continuous and select the appropriate summary statistics
- Calculate and interpret measures of centre and spread, and recognise when each is appropriate
- Select and construct an appropriate visualisation for a given variable type and data distribution
- State the components of a hypothesis test, interpret a p-value, and construct and interpret a confidence interval

Why revise this now?

**“Not everything that can be counted counts,
and not everything that counts can be measured.”**

— attributed Einstein

- Everything in Sections 1–3 of this module depends on these foundations
- A wrong choice of summary statistic or test costs marks in the exam
- The Irish weather data will be our running example throughout

What kind of data do we have?

Data types drive everything

“The method follows the measurement.”

The type of variable determines:

- What you can **calculate** (mean? median? count?)
- What **plot** is appropriate (histogram? bar chart?)
- What **model** to fit (regression? logistic? chi-squared?)

Warning

Wrong data type → meaningless output.

This is not a technicality: it's a fundamental error.

Data types

Qualitative (Categorical)

 **Nominal:** categories, no natural order
e.g. station name, blood type, country

 **Ordinal:** natural order, unequal gaps
e.g. degree class, Likert scale

Quantitative (Numeric)

 **Interval:** equal gaps, no true zero
e.g. temperature in °C, year

 **Ratio:** equal gaps, true zero = absence
e.g. rainfall (mm), height, income

Nominal

Categories with **no natural order**. You can count them and compute percentages, but you cannot rank or subtract.

Examples

- Weather station (Shannon, Dublin, Cork...)
- Blood type (A, B, AB, O)
- Country of birth
- Favourite colour

What you can do

- ✓ Count: how many in each category?
- ✓ Percentage: what proportion?
- ✗ Mean: “the average station is 2.3” = meaningless
- ✗ Rank: is Cork “more” than Dublin?

Ordinal

A **natural order** exists, but the **gaps between categories are not equal**. Ranking is valid; arithmetic is not.

Example: Spice level

Level	Rank
Mild	1
Medium	2
Hot	3
Very hot	4

What you can do

- ✓ Rank: Very hot > Hot > Medium
- ✓ Median: find the middle rank
- ✗ Mean: “the average spice level is 2.3”
- ✗ Subtract: “Very hot is twice as spicy as Medium”

The gap between Mild and Medium is not necessarily the same as between Hot and Very hot.

The Likert scale debate

Fatigue scale: 1 = No fatigue → 7 = Exhausted

Version 1: plain

1	2	3	4	5	6	7
None						Exhausted

Is the gap from 1 → 2 the same as from 6 → 7?

Version 2: labelled

Score	Label
1	No fatigue
2	Mild
3	Better than normal
4	Normal
5	Worse than normal
6	Very tired
7	Exhausted

 Warning

Are 3/4/5 equally spaced? Debatable.

Interval

Equal gaps between units, but **no true zero**. Zero is arbitrary, not an absence of the thing being measured.

Classic example: °C

- The gap from 10°C to 11°C equals the gap from 20°C to 21°C ✓
- But 0°C does not mean “no temperature”. It’s the freezing point of water
- 20°C is **not** twice as hot as 10°C ✗

Other examples

- Year (2020 is not “twice” 1010)
- IQ scores (zero does not mean zero intelligence)
- Dates and times

Ratios of interval values are meaningless. Differences are fine.

Ratio

Equal gaps and a **true zero**. Zero means a genuine absence of the quantity. All arithmetic is valid.

Examples

- Rainfall `prcp_mm`. 0mm = no rain ✓
- Number of rainy days in a month
- Height, weight, income
- Temperature in Kelvin
- Reaction time (seconds)

What this unlocks

- ✓ Mean and SD
- ✓ Ratios: Cork gets twice the rainfall of Dublin
- ✓ All the summary statistics and models in this module

Note

Most continuous variables in practice are ratio level.

Classify the Irish weather variables

Variable	Description	Type	What we can do
<code>station</code>	Weather station name	Nominal	Count, percentage, <code>table()</code>
<code>date</code>	Date of observation	Interval / ordered	Differences (days between), not ratios
<code>tmax_c</code>	Maximum daily temperature (°C)	Ratio / continuous	Mean, SD, all arithmetic
<code>tmin_c</code>	Minimum daily temperature (°C)	Ratio / continuous	Mean, SD, all arithmetic
<code>prcp_mm</code>	Daily rainfall (mm)	Ratio / continuous	Mean, median, ratios valid

Discussion: What can we say about stations? What can we say about temperatures?

Continuous vs discrete

Continuous

Can take any value within a range, including fractions.

- `tmax_c` = 14.7°C, 14.71°C, 14.712°C...
- Height, weight, rainfall

→ Use **histogram** to show distribution

→ Use **mean/SD or median/IQR** to summarise

Discrete

Takes only countable, distinct values.

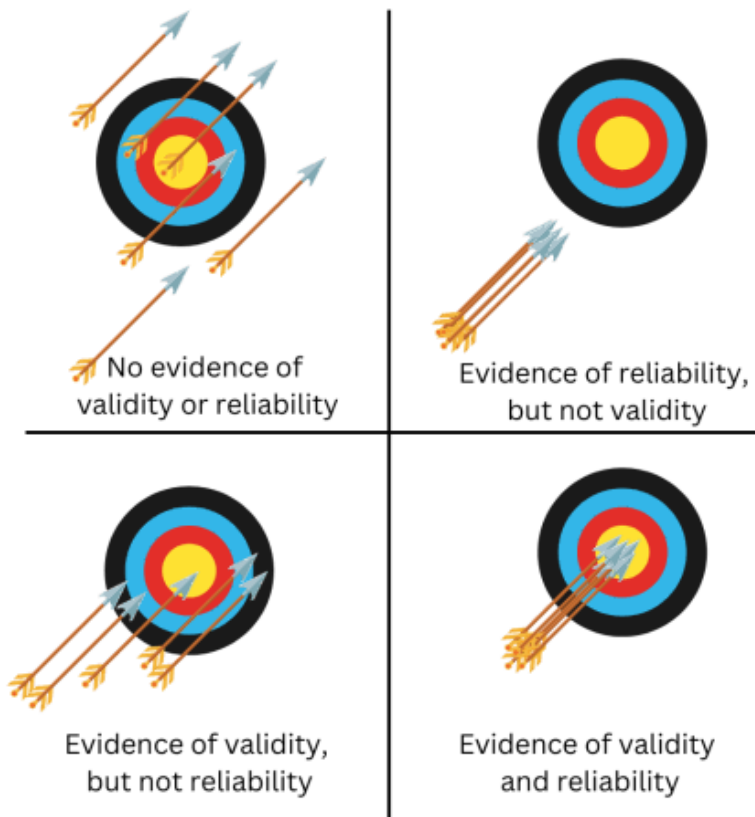
- Number of rainy days in a month: 0, 1, 2... 31
- Number of students in a lecture
- Number of accidents per week

→ Can also be **ratio level**

→ Bar chart or histogram (wide bins) both work

The distinction matters most for choosing plots and in later weeks, for choosing between regression and Poisson/negative binomial models.

Validity & reliability



Validity: refers to the credibility or believability of the research.

Does the data measure what it's supposed to?

Reliability: is the degree to which measurements provide stable and consistent results.

Would you get the same answer if you measured again?

Note

Example: A scale that always reads 2 kg too heavy is **reliable** (consistent) but **not valid** (wrong answer).

No statistical model can rescue a bad measurement.

Describing what we have

The three questions to answer first

1. What is the **shape**?
2. Where is the **centre**?
3. How **spread out** is it?

Warning

Never report a mean without first describing the shape.

A mean on a skewed distribution is misleading. It will not represent the typical value.

Categorical variables: Counts and percentages

```
1 table(ireland$station)
```

Cork	Dublin	Malin Head	Shannon	Valentia
7573	8675	8697	8555	8690

```
1 prop.table(table(ireland$station))
```

Cork	Dublin	Malin Head	Shannon	Valentia
0.1794975	0.2056174	0.2061389	0.2027732	0.2059730

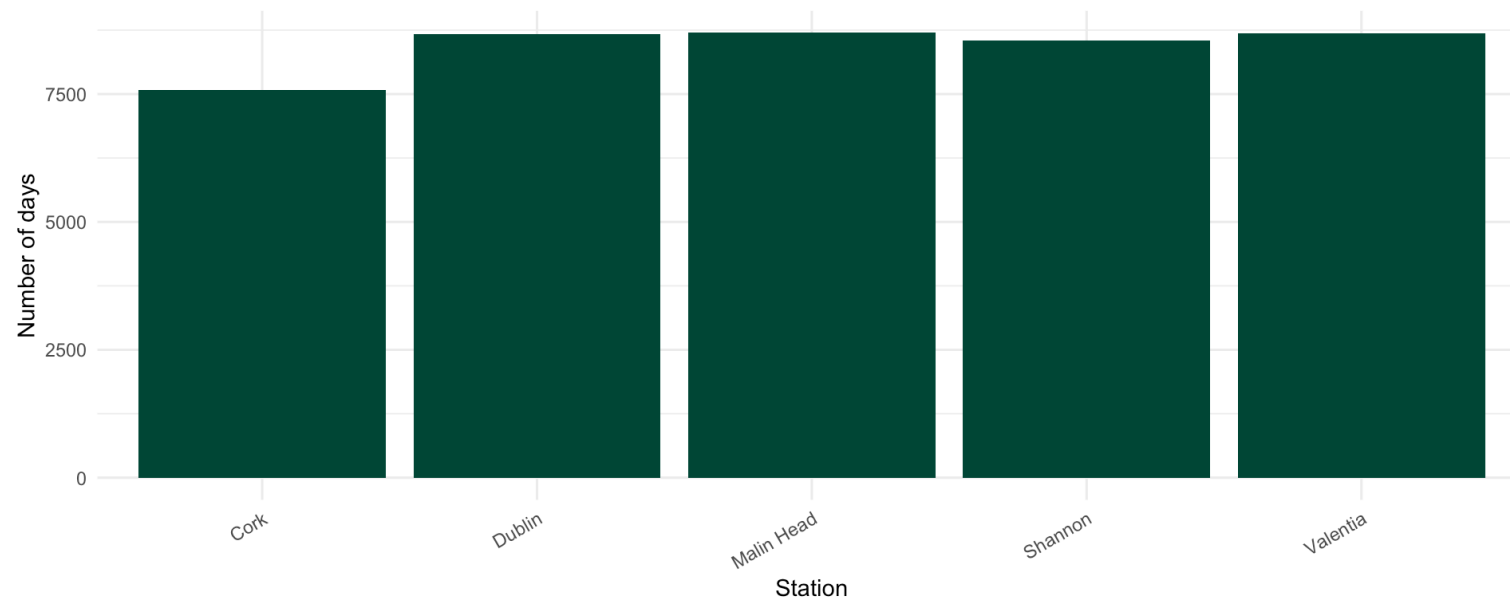
Visualising categorical data: Bar chart

```
1 ggplot(ireland, aes(x = station)) +  
2   geom_bar(fill = "#034638") +  
3   labs(  
4     x = "Station",  
5     y = "Number of days",  
6   ) +  
7   theme_minimal() +  
8   theme(axis.text.x = element_text(  
9     angle = 30, hjust = 1))
```

One bar per category.

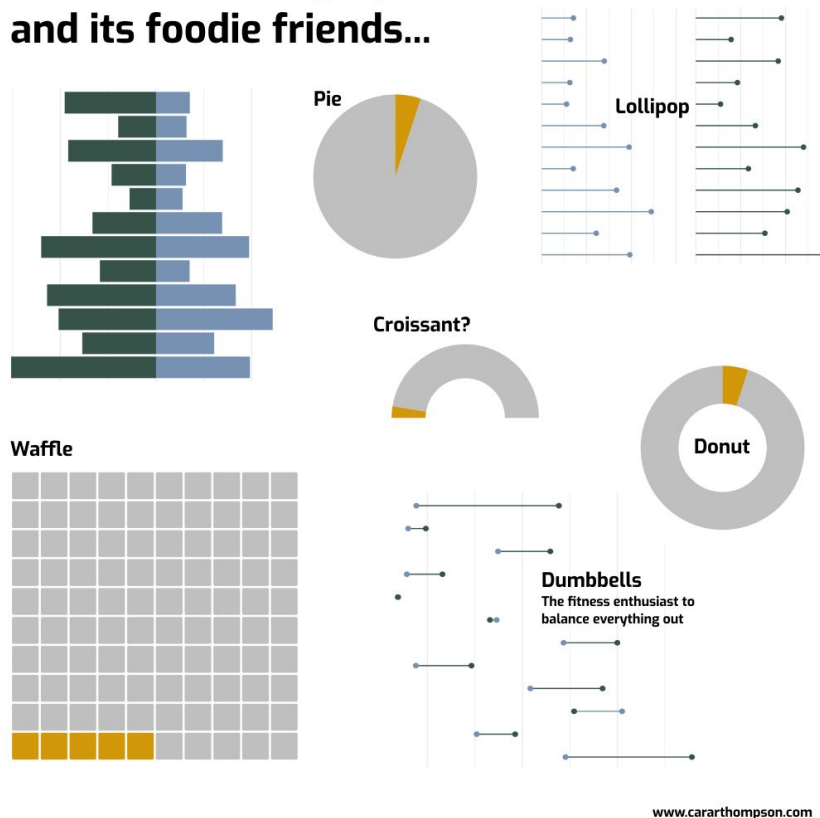
Height = count.

Do **not** use pie charts for more than 3-4 categories



Alternatives to the bar chart

The humble bar graph and its foodie friends...



The bar chart is a solid default, but not the only option for categorical data.

Lollipop: less visual clutter for many categories; the dot draws the eye to the value

Waffle: each square = one unit or %; good for part-of-a-whole when you want to avoid pie

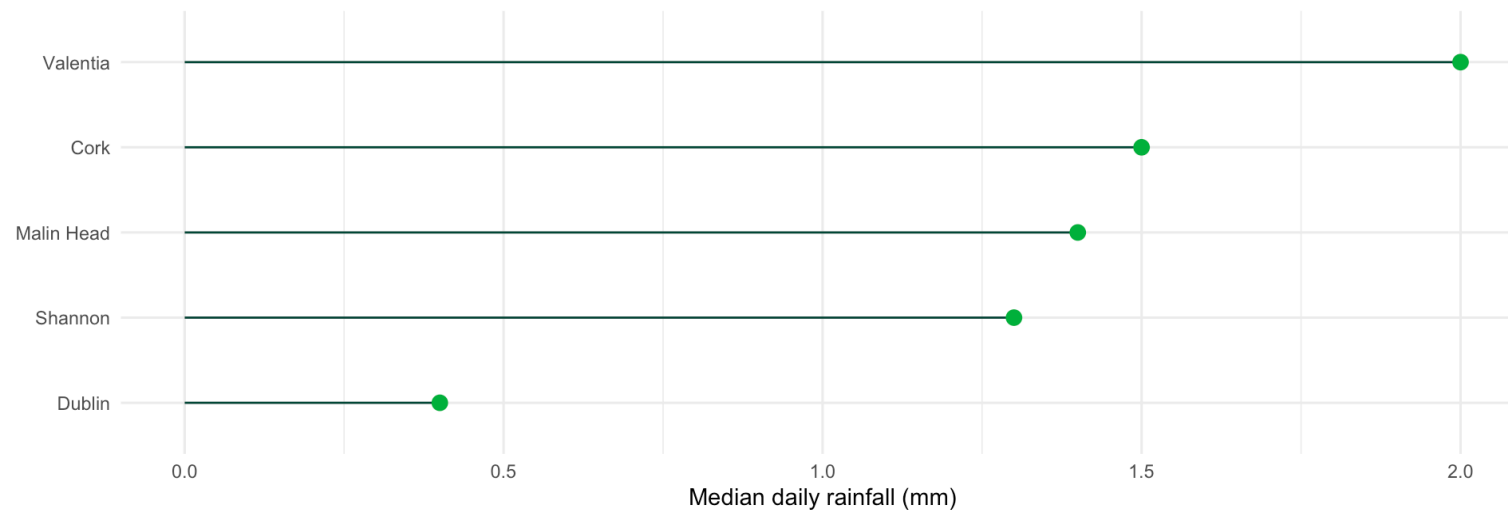
Donut / Pie: only when ≤ 4 categories and slices genuinely sum to 100%

Dumbbell: two values for the same category (e.g. before/after, male/female); comparison is built in

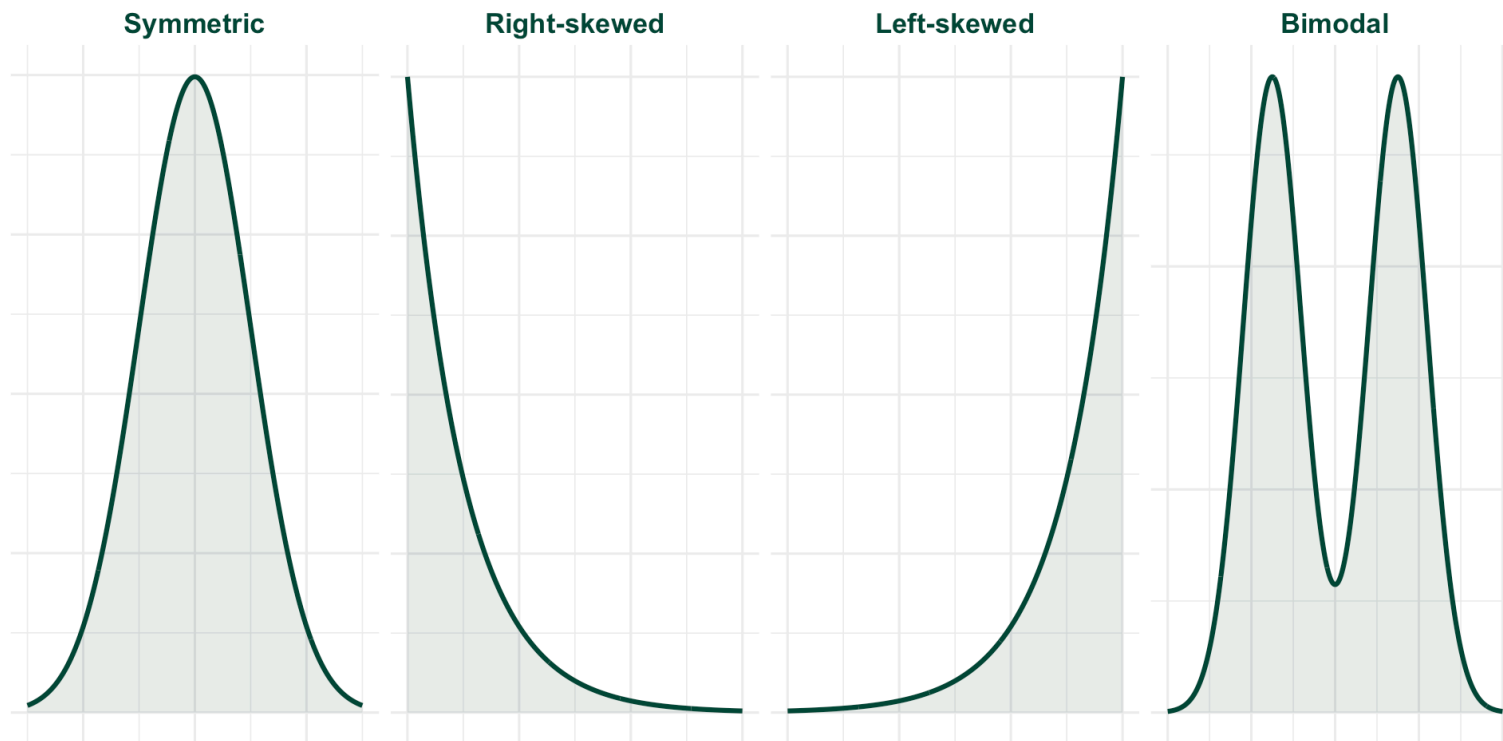

```

1 rain <- tapply(ireland$prcp_mm,
2               ireland$station,
3               median, na.rm = TRUE)
4 df <- data.frame(station = names(rain),
5                 median_rain = rain)
6 ggplot(df, aes(x = median_rain,
7               y = reorder(station,
8                 median_rain))) +
9   geom_segment(aes(x = 0, xend = median_rain,
10                  yend = station),
11               colour = "#034638") +
12   geom_point(colour = "#00B140", size = 3) +
13   labs(x = "Median daily rainfall (mm)",
14        y = NULL) +
15   theme_minimal()

```



The shape of continuous data



Always plot a histogram first.

Shape determines which summary statistics to report.

Mean

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

```
1 mean(ireland$prcp_mm, na.rm = TRUE)
[1] 3.577084
```

The **balancing point** of the distribution.

Sensitive to extreme values: one very wet day pulls the mean up.

✓ Appropriate when the distribution is **symmetric**

✗ Misleading when the distribution is **skewed** or has **outliers**

→ For **prcp_mm** (right-skewed): the mean overstates the typical day's rainfall

→ For **tmax_c** (roughly symmetric): the mean is a reasonable summary

i Note

Always use **na.rm = TRUE** : weather data has missing observations.

Median

The **middle value** when all observations are sorted in order.

```
1 median(ireland$prcp_mm, na.rm = TRUE)
[1] 1.2
```

If **n** is even: average of the two middle values.

Robust to outliers: one extremely wet day barely moves the median.

- ✓ Appropriate when the distribution is **skewed**
- ✓ Appropriate for **ordinal** data (e.g. Likert)
- ✓ Appropriate when **outliers** are present
- ✗ Less efficient than the mean when the distribution really is symmetric (wastes information)

Mean vs median: Which to report?

Symmetric distribution

```
1 mean(ireland$tmax_c, na.rm = TRUE)
[1] 13.72133
```

```
1 median(ireland$tmax_c, na.rm = TRUE)
[1] 13.7
```

Mean $\approx$ median $\rightarrow$ report **mean + SD**

tmax_c is roughly symmetric, so mean is appropriate

Skewed distribution

```
1 mean(ireland$prcp_mm, na.rm = TRUE)
[1] 3.577084
```

```
1 median(ireland$prcp_mm, na.rm = TRUE)
[1] 1.2
```

Mean $\gg$ median $\rightarrow$ report **median + IQR**

prcp_mm is right-skewed, so mean is misleading

Note

General rule: Symmetric $\rightarrow$ mean + SD. Skewed or outliers $\rightarrow$ median + IQR. Categorical $\rightarrow$ count + %

Standard deviation

The average distance from the mean.

$$s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$$

```
1 sd(ireland$tmax_c, na.rm = TRUE)
[1] 4.462409
```

- ✓ Pair with the **mean**
- ✓ Same units as data, easy to interpret
- ✗ Sensitive to outliers, one extreme value inflates it
- ✗ Do **not** pair with the median

→ A daily max temperature SD of ~4°C means most days are within ~4°C of the mean

i Note

We use $n - 1$ (not n) in the denominator. This gives an unbiased estimate of the population SD.

Variance

$$s^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$$

```
1 var(ireland$tmax_c, na.rm = TRUE)
```

```
[1] 19.91309
```

Variance = SD². Units are **squared** (e.g. °C²)

Less interpretable on its own.

Note

Variance is central to the maths; SD is what you report to people.

Why care about variance?

- Underpins **all regression theory** in this module
- The “noise” term in every model is variance
- ANOVA literally partitions variance into explained and unexplained components

IQR and the five-number summary

```
1 IQR(ireland$prcp_mm, na.rm = TRUE)
```

```
[1] 4.8
```

```
1 fivenum(ireland$prcp_mm)
```

```
[1] 0.0 0.0 1.2 4.8 150.4
```

```
1 summary(ireland$prcp_mm)
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.	NAs
0.000	0.000	1.200	3.577	4.800	150.400	5861

Five-number summary: Min, Q1, Median, Q3, Max

$\text{IQR} = \text{Q3} - \text{Q1} = \text{middle 50\% of the data}$

✓ Pair with the **median**

✓ **Robust to outliers** .Q1 and Q3 barely move when extremes change

→ IQR contains the middle 50% of observations

→ Values beyond $\text{Q1} - 1.5 \times \text{IQR}$ or $\text{Q3} + 1.5 \times \text{IQR}$ are flagged as potential outliers in a boxplot

Robust vs non-robust statistics

Statistic	Robust?	Why
Median	✓ Yes	Adding one extreme observation barely changes it
IQR	✓ Yes	Based on Q1 and Q3, not affected by tails
Mean	✗ No	One very large value pulls it upward
SD	✗ No	Deviations are squared, outliers inflate it

```
1 c(mean = mean(ireland$prcp_mm, na.rm = TRUE),  
2   median = median(ireland$prcp_mm, na.rm = TRUE))
```

```
      mean  median  
3.577084 1.200000
```

For Irish rainfall: mean >> median. The right tail of heavy rain days drags the mean up.

The rule in practice

Step 1: Plot a histogram

Step 2: Assess the shape

Step 3: Choose your summary statistics



Symmetric

Report **mean + SD**



Skewed or outliers

Report **median + IQR**



Categorical

Report **count + %**

 **Note**

Always plot before you summarise.

Describing the Irish weather data

`tapply(x, INDEX, FUN)` applies a function to each group separately.

```
1 # Mean max temperature by station
2 tapply(ireland$tmax_c, ireland$station, mean, na.rm = TRUE)
```

Cork	Dublin	Malin Head	Shannon	Valentia
13.65741	14.19320	12.63870	14.54269	13.92582

```
1 # Median daily rainfall by station
2 tapply(ireland$prcp_mm, ireland$station, median, na.rm = TRUE)
```

Cork	Dublin	Malin Head	Shannon	Valentia
1.5	0.4	1.4	1.3	2.0

- Which station is **warmest** on average? (Cork or Valentia)
- Which station has the **highest median daily rainfall**? (Valentia = exposed Atlantic coast)
- Which is **driest**? (Dublin = in the rain shadow of the mountains)

summary() in R

```
1 summary(ireland[, c("tmax_c", "tmin_c", "prcp_mm")])
```

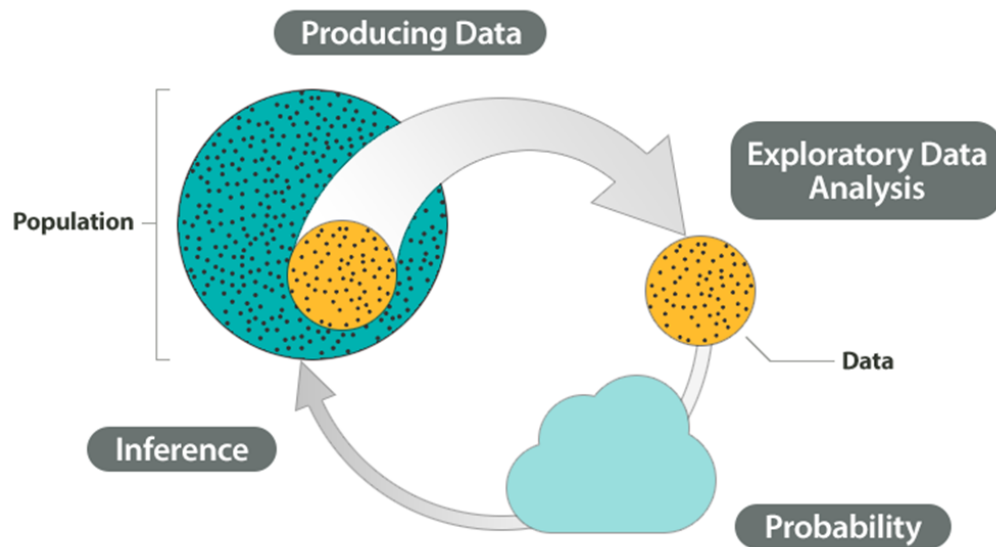
tmax_c	tmin_c	prcp_mm
Min. : -4.70	Min. : -12.200	Min. : 0.000
1st Qu.: 10.40	1st Qu.: 4.400	1st Qu.: 0.000
Median : 13.70	Median : 7.700	Median : 1.200
Mean : 13.72	Mean : 7.474	Mean : 3.577
3rd Qu.: 17.00	3rd Qu.: 10.800	3rd Qu.: 4.800
Max. : 33.00	Max. : 20.500	Max. : 150.400
NAs : 10264	NAs : 11121	NAs : 5861

summary() returns the five-number summary + mean for each numeric variable, plus NA counts.

- **Min / Max:** check for implausible values
- **1st Qu. / 3rd Qu.:** IQR = Q3 – Q1
- **Median vs Mean:** if they differ substantially: skewed distribution
- **NA's:** how many missing values? Could this bias the analysis?

From data to conclusions

Populations and samples



We **never** observe the full population.

We use **sample statistics** to estimate **population parameters**:

Sample	Population
$\bar{x}$ (mean)	μ (mu)
s (SD)	σ (sigma)
$\hat{p}$ (proportion)	p

Quality of estimate depends on:

- **Sample size** n (larger $\rightarrow$ more precise)
- **Representativeness** (random sample $\rightarrow$ unbiased)

Types of study

Observational vs Experimental

 **Observational:** we record what happens without intervening. The Irish weather dataset is observational.

 **Experimental:** we randomly assign treatments and control conditions. Allows causal claims.

Cross-sectional vs Longitudinal

 **Cross-sectional:** one snapshot in time. e.g. a single survey.

 **Longitudinal:** repeated measures over time. The Irish weather data: 2000–2023, the same stations every year.

Note

The Irish weather data is **observational** and **longitudinal**.

Correlation $\neq$ causation

A correlation tells you:

✓ There is a **statistical relationship**

✗ NOT why it exists

✗ NOT that one **causes** the other

Common explanations for correlation without causation:

- **Confounding** variable driving both
- **Coincidence** (especially with small n)
- **Reverse causation**

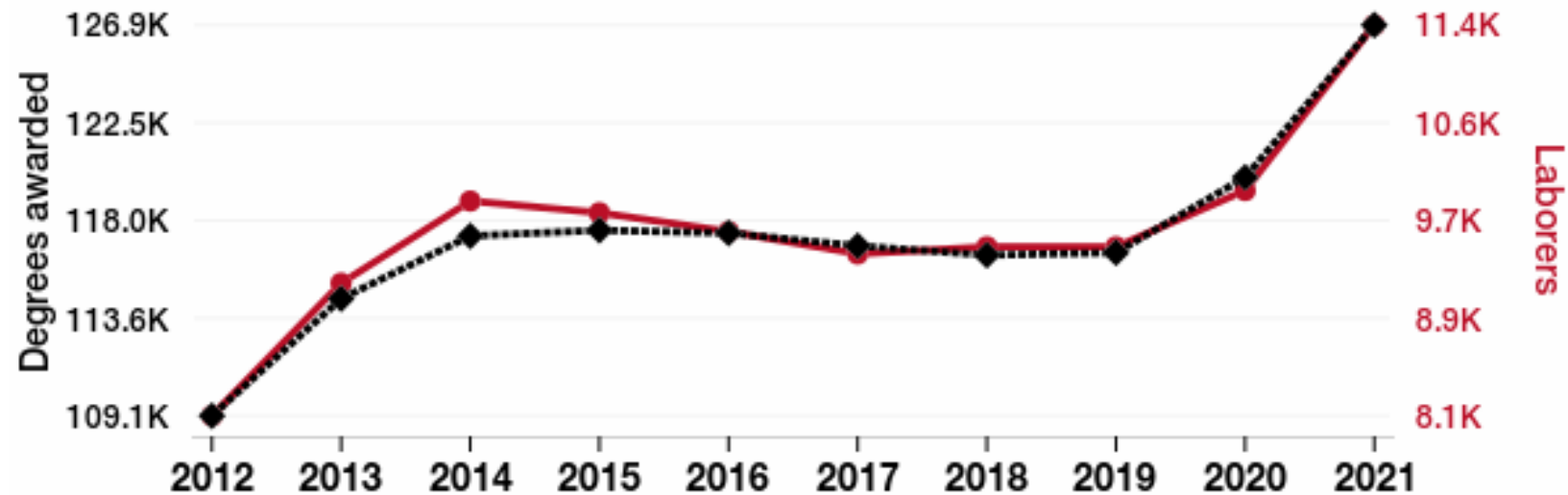
Warning

We can establish **association** with observational data. **Causation** requires experimental design.

Bachelor's degrees awarded in Psychology

correlates with

The number of groundskeepers in Utah



◆ Bachelor's degrees conferred by postsecondary institutions, in field of study: Psychology · Source: National Center for Education Statistics

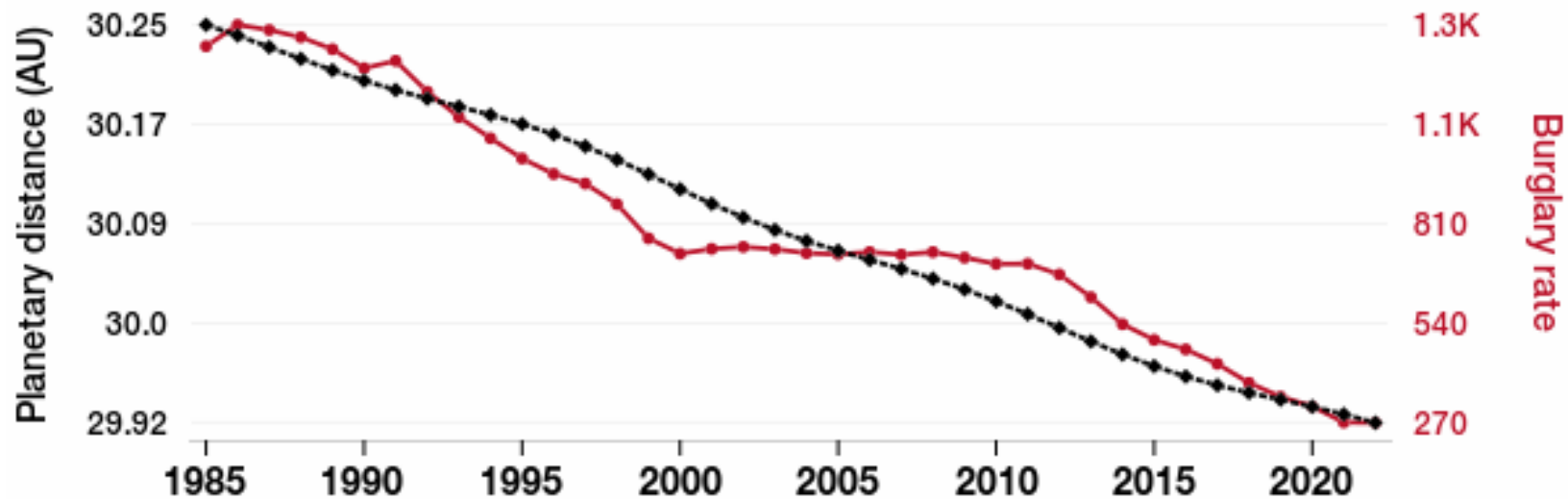
● BLS estimate of landscaping and groundskeeping workers in Utah · Source: Bureau of Labor Statistics

2012-2021, $r=0.990$, $r^2=0.980$, $p<0.01$ · tylervigen.com/spurious/correlation/1781

The distance between Neptune and the Sun

correlates with

Burglary rates in the US



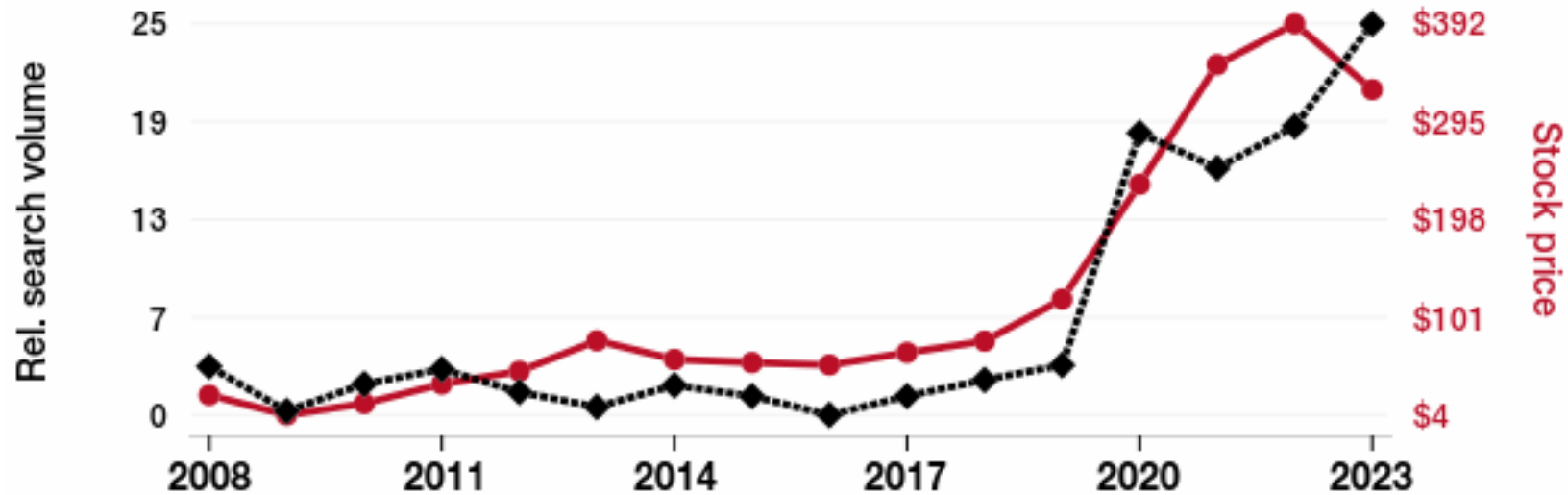
- ◆--- The average distance between Neptune and the Sun as measured on the first day of each month · Source: Cacculated using Astropy
- The burglary rate per 100,000 residents in United States · Source: FBI Criminal Justice Information Services

1985-2022, $r=0.969$, $r^2=0.940$, $p<0.01$ · tylervigen.com/spurious/correlation/1035

Google searches for 'where can i stream friends'

correlates with

Lululemon's stock price (LULU)



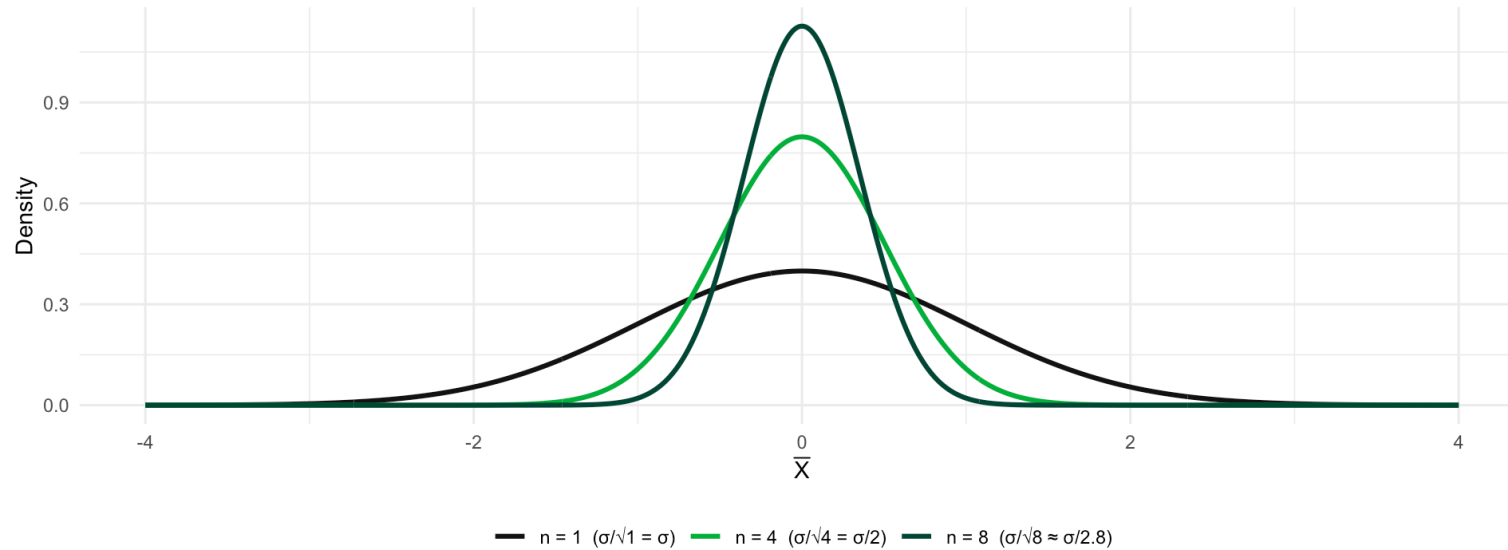
◆--- Relative volume of Google searches for 'where can i stream friends' (Worldwide, without quotes) · Source: Google Trends

●— Opening price of Lululemon Athletica (LULU) on the first trading day of the year · Source: LSEG Analytics (Refinitiv)

2008-2023, $r=0.923$, $r^2=0.852$, $p<0.01$ · tylervigen.com/spurious/correlation/2694

Sampling distribution of $\bar{X}$

Sampling distribution of $\bar{X}$ narrows as n increases

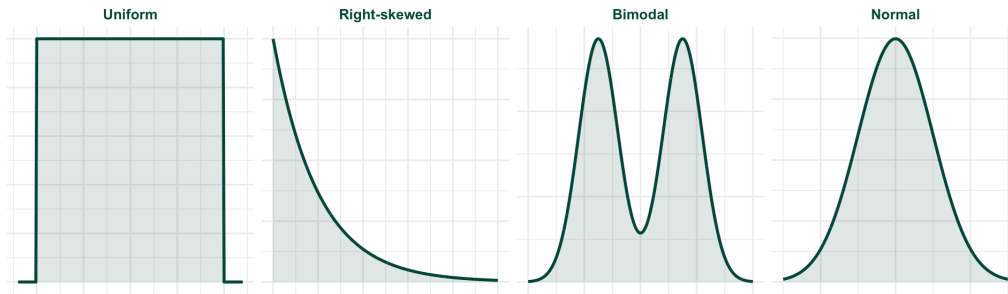


$$\text{If } X \sim N(\mu, \sigma^2), \text{ then } \bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$$

Larger $n \rightarrow$ narrower distribution $\rightarrow$ more precise estimates.

The Central Limit Theorem (CLT)

Any population shape $\rightarrow \bar{X}$ becomes Normal (CLT)



For a random sample from **any** population:

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \xrightarrow{n \rightarrow \infty} N(0, 1)$$

What this buys us:

Even if the population is skewed or bimodal, the sampling distribution of $\bar{X}$ becomes approximately normal for large n .

This is why we can use the t-test on non-normal data provided n is large enough (rule of thumb: $n \geq 30$).

Note

The CLT is what makes most of classical statistics work.

95% Confidence interval for μ

From the CLT:

$$\bar{x} \pm \frac{1.96 \sigma}{\sqrt{n}}$$

When σ unknown (always in practice), use s . For small n (< 30), replace 1.96 with the t_{n-1} critical value:

$$\bar{x} \pm \frac{t_{\alpha/2, n-1} s}{\sqrt{n}}$$

Warning

Common misinterpretation:

 “There is a 95% probability that μ is in this interval”

 “If we repeated this procedure many times, 95% of the intervals constructed would contain μ ”

The interval either contains μ or it doesn't. There is no probability about it once the data are observed.

p-values

Note

p-value: The probability of observing data this extreme or more, **given that H_0 is true.**

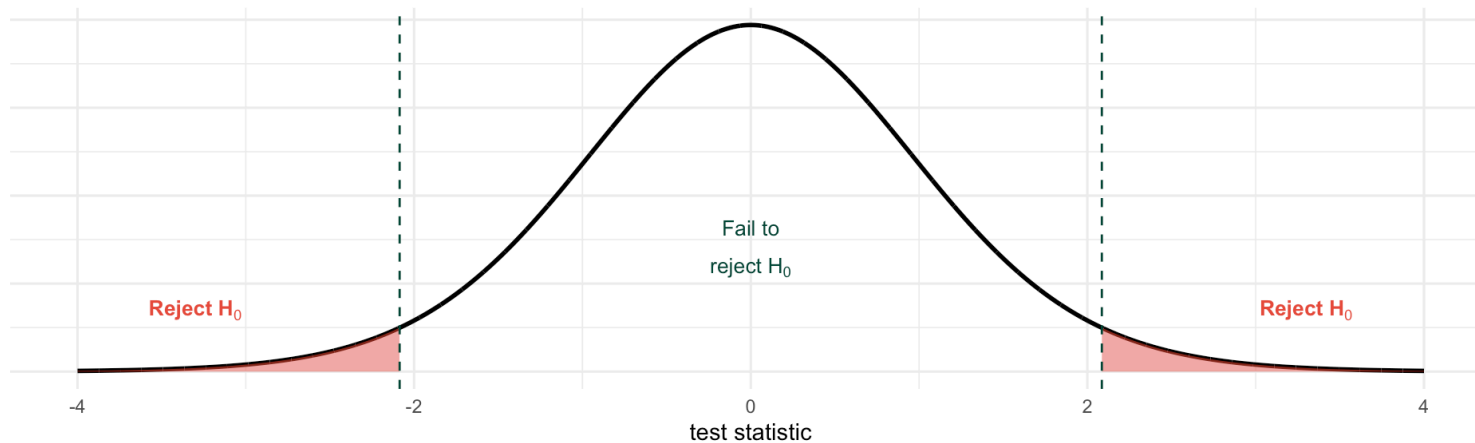
- $p < 0.05 \rightarrow$ reject $H_0 \rightarrow$ statistically significant
- $p \geq 0.05 \rightarrow$ fail to reject H_0

 **Fail to reject $\neq$ prove H_0 is true.**

Absence of evidence is not evidence of absence.

Steps for hypothesis testing

1. State H_0 and H_a (the null and alternative)
2. State α : usually 5% (= 0.05)
3. Compute the **test statistic**
4. Find the **p-value**
5. Decision: if $p < \alpha$, reject H_0 in favour of H_a
6. **Conclusion in context**: always state what it means for your data, not just “ $p < 0.05$ ”



One-sample t-test

Test whether a population mean equals a specified value μ_0 .

$$T = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \sim t_{n-1}$$

Question: Is the mean daily max temperature in Shannon significantly different from 14.54°C?

```
1 shannon <- ireland$tmax_c[ireland$station == "Shannon"]
2 t.test(shannon, mu = 14.54)
```

One Sample t-test

```
data: shannon
t = 0.040663, df = 5005, p-value = 0.9676
alternative hypothesis: true mean is not equal to 14.54
95 percent confidence interval:
 14.41306 14.67232
sample estimates:
mean of x
 14.54269
```

Reading the output:

- **t =** the test statistic
- **df =** degrees of freedom ($n - 1$)
- **p-value =** evidence against H_0
- **95 percent confidence interval:**
plausible range for μ
- **mean of x** the sample mean $\bar{x}$

Decision rule: if **p-value < 0.05**, reject $H_0 : \mu = 14.54$.

Two-sample t-test

Test whether two population means differ.

$$T = \frac{\bar{x}_1 - \bar{x}_2}{SE} \sim t_{n_1+n_2-2}$$

Signal = difference between means

Noise = pooled standard error

```
1 t.test(tmax_c ~ station,
2       data = subset(ireland,
3                     station %in% c("Cork", "Dublin")),
4       var.equal = TRUE)
```

Two Sample t-test

```
data:  tmax_c by station
t = -6.1749, df = 13205, p-value = 6.815e-10
alternative hypothesis: true difference in means between group
Cork and group Dublin is not equal to 0
95 percent confidence interval:
 -0.7058702 -0.3657103
sample estimates:
 mean in group Cork mean in group Dublin
      13.65741      14.19320
```

Key syntax:

y ~ group formula notation: “y by group”

var.equal = TRUE assumes equal variance in both groups (pooled t-test)

Reading the output:

- **t =** and **p-value =** as before
- **95 percent confidence interval:** for the **difference in means**
- **mean in group Cork** vs **mean in group Dublin**: the two group means

Decision: Is there sufficient evidence that Cork and Dublin differ in mean max temperature?

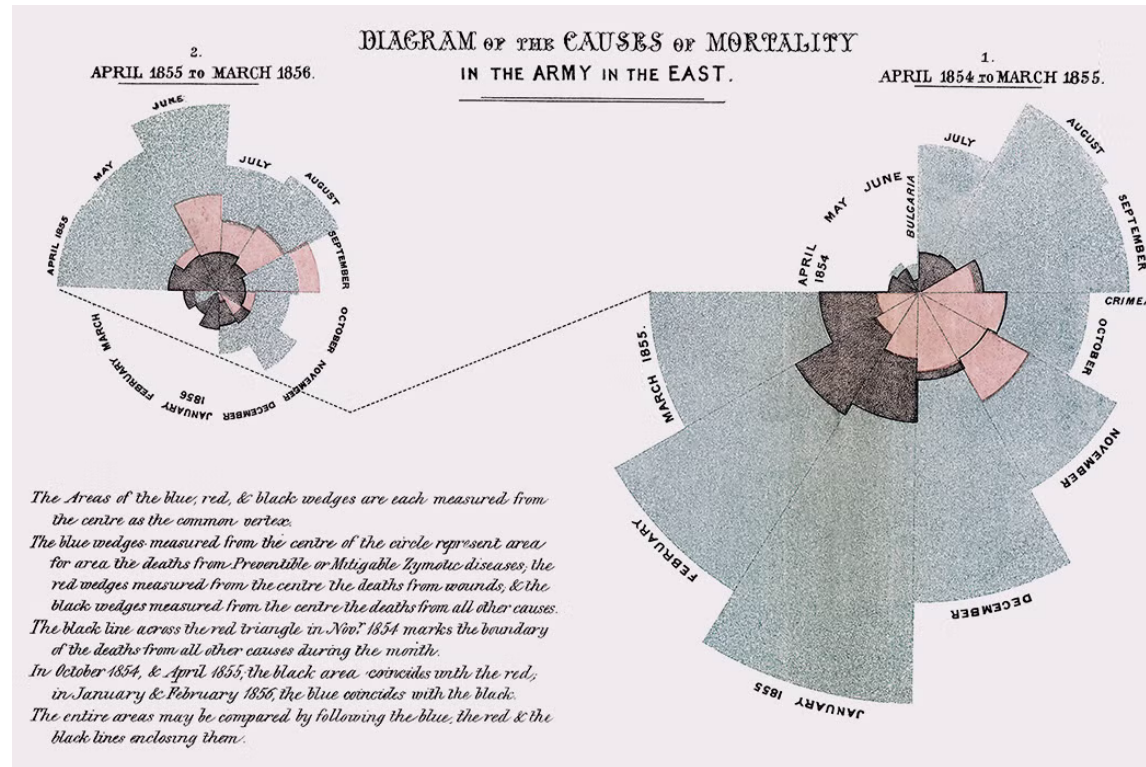
All graphs are comparisons

Every plot asks “compared to what?”

The label, axis, and context determine whether the comparison is honest.

“Never display a graph you can’t talk about.” Gelman & Hill (ROS)

Why visualise?

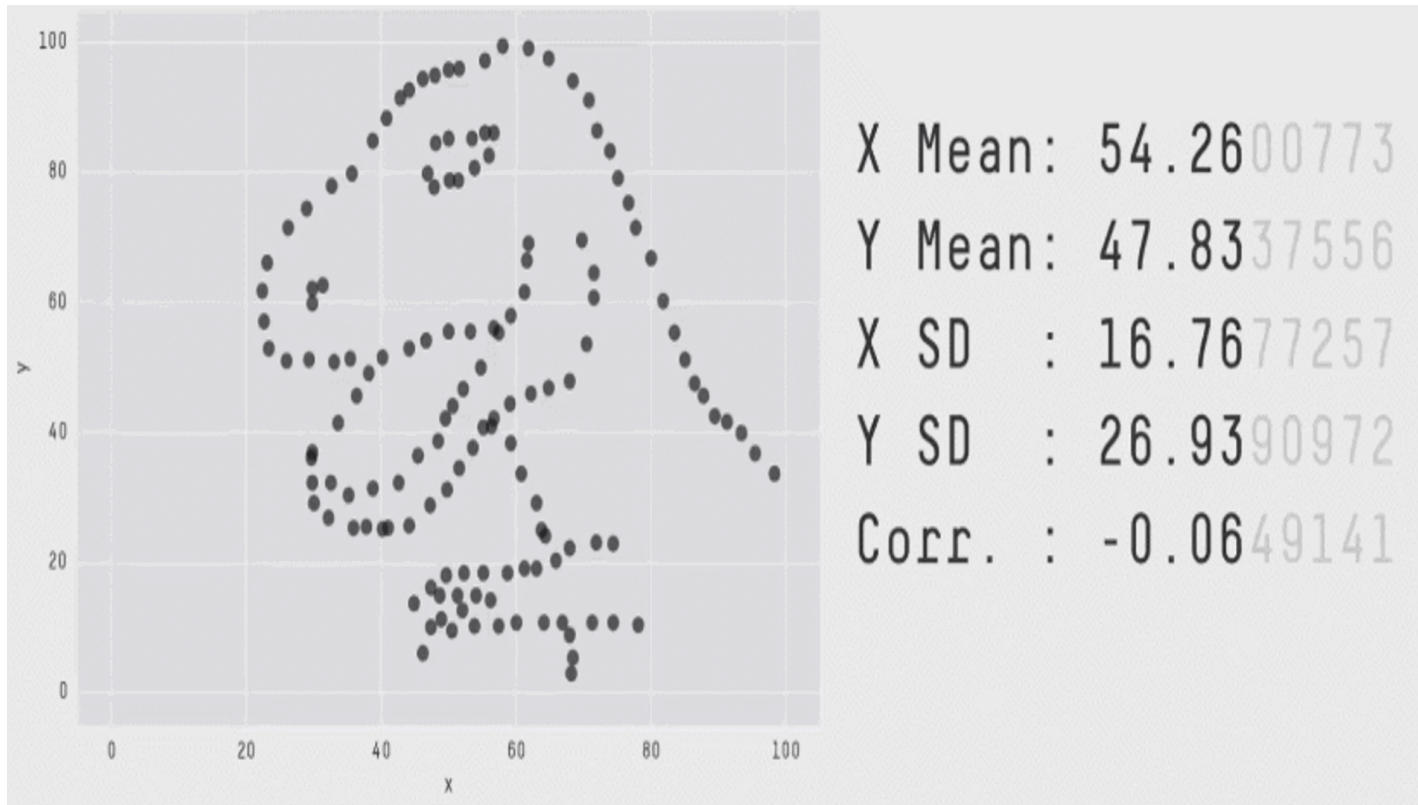


Florence Nightingale's 1858 polar area chart showed that more British soldiers in the Crimean War died of **preventable disease** than combat wounds. The chart changed military medical practice.

A single plot can make an argument that a table of numbers cannot.

Warning!

The following sequence of scatter-plots all have the same statistics (to 2 decimal places) and yet they show a variety of interesting patterns:



Choosing the right plot

Question	Variable type	Plot
How are categories distributed?	Nominal	Bar chart
What is the shape of a continuous variable?	Continuous	Histogram
Where is the centre and spread?	Continuous	Boxplot
Is there a relationship between two variables?	Two continuous	Scatter plot
How does a continuous variable differ by group?	Continuous + categorical	Side-by-side boxplot

Every plot implies a comparison. Always ask: *compared to what?*

ggplot2: The pattern

```
1 # Every ggplot2 plot has the same skeleton:
2 ggplot(data = <DATA>, aes(x = <X>, y = <Y>)) +
3   geom_<TYPE>() +
4   labs(x = "<x label>", y = "<y label>",
5         title = "<title>") +
6   theme_minimal()
```



ggplot(data,
aes(...))

What to plot and **which**
variable goes where



geom_*()

How to draw it

geom_bar ,
geom_histogram ,
geom_boxplot , **geom_point**



labs() + **theme_*()**

Labels and **appearance**

Always label axes and give a
title

R functions

Function	Purpose
<code>table(x)</code>	Frequency table for a categorical variable
<code>mean(x) / median(x)</code>	Measures of centre
<code>sd(x) / var(x) / IQR(x)</code>	Measures of spread
<code>hist(x)</code>	Histogram of a continuous variable
<code>boxplot(y ~ group, data = df)</code>	Comparative boxplot by group
<code>plot(x, y)</code>	Scatter plot of two continuous variables
<code>t.test(x, y)</code>	Two-sample t-test
<code>t.test(x, mu = value)</code>	One-sample t-test

Summary

- 🔧 **Variable type** determines what you can calculate, what plot to use, and what model to fit
- 📊 **Plot first, summarise second:** shape determines mean+SD vs median+IQR
- 📈 **All graphs are comparisons:** always ask what is being compared and whether it's honest
- 🔬 **CLT** justifies the use of t-tests and confidence intervals even when the population is not normal
- 🔵 **p-value** is evidence against H_0 . Not proof of H_a , and not a measure of effect size